

ACCIDENTS FOLLOWING VACCINATION.

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THIS study is suggested because of the increasing difficulty in securing compliance with school requirements in reference to vaccination, due to the activity of anti-vaccinationists in exaggerating both the number and character of these unfortunate accidents.

Cancer, consumption, typhoid fever, ulcerated stomach, leprosy, diphtheria, syphilis and many other diseases have been charged to vaccination, but these require no discussion here. Several cases of cerebro-spinal meningitis have been reported as following vaccination, but here again no causal relation has been established. Septicemia in various forms, due to invasion of pathogenic organisms at some stage of the vaccinal process from accident or carelessness is occasionally noted.

Recently a Philadelphia court awarded a negro \$5,000 damages for the loss of a leg, said to be due to septicemia, following vaccination by a city physician in 1910. Carelessness was alleged. The verdict, however, has been set aside by a higher court.

A case is now pending in Chestertown, Md., where the commissioners are sued for \$10,000 for the death of a negro child through alleged carelessness in vaccination by the county physician.

Recently, three cases of anterior poliomyelitis following vaccination in

from ten to twenty days, have been reported from New York State. So far as is known, these children did not come in contact with each other or with other cases of poliomyelitis, so that the parents naturally concluded that vaccination was at least a factor in some way.

While cases of the classes enumerated are a drawback to enforcing vaccination, they are insignificant as compared with tetanus, and it is in a study of this disease in its relation to vaccination that we are chiefly interested.

I have been able to secure more or less complete histories of 82 cases occurring in the United States and Canada since 1899. Some of these, prior to 1902, were discussed by Wilson,* and are taken from his paper; some more recent Philadelphia cases are taken from Wadsworth's† report of coroner's cases and from records of the coroner's court. I wish here to express to Doctor Wadsworth my appreciation of his assistance in securing copies of these records.

First analyzing these somewhat in detail we note the following facts.

Case 1. Pushed against door by playmate at school. Scab torn off. Tetanus developed in 24 hours.

Case 2. Large, deep ulcer present 20 days after vaccination. Arm dirty.

*An analysis of fifty-two cases of tetanus following vaccination with reference to source of infection. *Journal of A. M. A.*, 5/3/02.

†Records of cases of tetanus in Philadelphia during recent years by Dr. Wm. S. Wadsworth. From the *Transactions of the College of Physicians of Philadelphia*, 1913.

Case 3. Dirty, sloughing ulcer. Child in poor health.

Case 4. Deep ulcer present.

Case 5. Sore about $1\frac{1}{2}$ inches in diameter filled with pus; partly covered with a dirty bandage.

Case 7. Deep ulcer present.

Case 8. Dressed at time of vaccination with cotton wadding from an old quilt. Scab knocked off.

Case 9. Scab torn off arm; sleeve and arm both dirty at time tetanus developed.

Case 10. Large ulcer present. Arm and sleeve filthy.

Case 13. Bad, sloughing sore.

Case 14. Large suppurating ulcer, exposed to clothing and atmosphere.

Case 15. Deep ulcer in the arm, covered with a greenish matter.

Case 16. Large suppurating ulcer, which the doctor believes was infected by the finger nails of the child scratching it.

Case 17. Dressed by mother; dirty dressing retained for days. *Recovery.*

Case 18. Child wore shield and "dirt got under it."

Case 19. Shield left in place for over three weeks; large, deep, ulcer present; dressed with flaxseed poultice and rag bandage.

Case 20. Large ulcer; scab removed and dressed at home. Child assisted in taking up potted plants. Cultures from earth near her home showed tetanus bacilli.

Case 22. Scab torn off; ulcer dressed at home; tetanus organisms in soil near child's residence.

Case 23. Celluloid shield fastened with adhesive strips; not removed for 28 days. Child slept with father who

had charge of the horses in the stable on floor below.

Case 24. Shield not removed for 21 days. Filled with discharge.

Case 25. Child's father was coachman and had charge of horses in stable on floor below. Child slept with father; vaccinated on the leg; ulcer size of a quarter.

Case 26. Bunion plaster used as shield.

Case 27. Dressed by mother; dirty dressing retained for days. *Recovery.*

Case 28. Bunion plaster applied when vaccinated.

Case 29. Muslin shield used; had not been removed and pus was escaping from shield.

Case 30. Child attended masquerade with wound exposed to red calico domino.

Case 31. Severe ulcer. Threw bandages on ground and then replaced them on the arm.

Case 32. Linen shield was used; had not been removed when tetanus set in.

Case 33. Muslin shield was used; ulcer about the size of a half dollar; later no covering used; lint from shirt ulcerated into wound.

Case 34. Ulcer about $1 \times \frac{3}{4}$ of an inch.

Case 35. Shield was removed in one week; ulcer about the size of a quarter.

Case 36. Vaccinated by a druggist.

Case 37. Surface to be vaccinated not washed; child played in lumber yard near horses.

Case 39. Patient frequently at work shoveling the dirt in yard. Dirt rich in tetanus bacilli; scratched with finger nails.

Case 40. Shield not removed for 22

days; filled with excretion which was running down the child's arm.

Case 41. Arm poulticed with plantain leaves, then with cow manure.

Case 42. Shield was filled with pus without changing for days; scab lost on 22d day and rag worn from that time.

Case 43. Vaccinated by policeman; severe ulcer; rag dressing; unclean surroundings.

Cases 45-6-7-8. Insane patients in the Philadelphia Hospital; filthy habits.

Case 49. Deep ulcer.

Case 53. Improperly cared for arm.

Case 54. Vaccinated on farm where a horse, cow and calf had previously died of tetanus.

Case 55. Played in yard; wound opened twice by falls.

Case 56. Physician's child; father believes case of secondary infection.

Case 59. Scab knocked off in school yard; onset of tetanus eight days later.

Case 60. A blow on arm October 4th; tetanus developed, followed by death on October 9th.

Case 62. Vaccine wound infected by scratching.

Case 65. Played about stable.

Case 71. Infected on farm.

Case 73. Sores on finger and ear.

Case 74. Subsequently ran splinter in the hand, which later became infected.

Case 75. Dressed with lettuce leaves once or twice daily.

Case 77. Blow received on the arm ten days after vaccination.

Case 78. Underclothing stuck to the sore.

Thus we note that out of 82 cases, we have histories of 54 showing possi-

bilities of subsequent infection of the vaccination wound; such as an improperly used shield, dirty dressings, traumatism, etc.; and a more or less intimate association with stables in several instances.

A further investigation shows that the disease was much longer in developing after vaccination than the usual incubation period of tetanus.

Developing within one week, 2....	2½ per cent.
Developing in from one to two weeks, 4.....	4¾ per cent.
Developing in from two to three weeks, 34.....	41½ per cent.
Developing in from three to four weeks, 17.....	20¾ per cent.
Developing in over four weeks, 6...	7¼ per cent.

All but four of these cases resulted in death; whereas in the 200 cases of tetanus reported by Andrews and Morgan in which the incubation period was fourteen days or over, 77 died; a mortality of only 38½ per cent. This would indicate that the infection was subsequent to vaccination. A further inspection of our list shows that of the 47 cases in which we have been able to obtain the number of days from the onset of tetanus to death, we find that 24, or slightly over 50 per cent. died in 48 hours, while 42, or something over 95 per cent. died within seven days.

Andrews and Morgan* in their table of 252 cases originally investigated show only 27 cases with an incubation period of over 14 days and of this number 19, or something over 70 per cent. recovered; while the average period between onset and death of the remaining eight was 6½ days.

* Tetanus. A preliminary report of a statistical study by James M. Andrews, M. D., LL. D., and Arthur C. Morgan, M. D., Philadelphia, Pa., Journal A. M. A. July 29, 1905.

CHART 1.

	Location.	Date of vaccination.	Date of onset of tetanus.	Nb. of days between vaccination and onset.	Date of death.	No. of days between onset and death.	Reported by
1.	Columbus, Ohio.....	1899		3 wk.			
2.	Cleveland, Ohio.....	1/3/99	1/24	21 da.	1/26	2	
3.	Omaha, Nebraska.....	1899			2/11		
4.	Brewster, N. Y.....	Oct. '99		21 da.			Wilson
5.	Chicago, Ill.....	10/3/99	10/16	13 da.	10/30	14	
6.	St. Louis, Mo.....				12/23		
7.	St. Louis, Mo.....	1900			10/8		
8.	Paris, Tenn.....	11/27/00		3 wk.	Death		Wilson
9.	Kalamazoo, Mich.....	1/3/01	1/27	24 da.	2/1	5	Wilson
10.	Cleveland, Ohio.....	1/3/01	1/24	21 da.	1/26	2	
11.	Chester, Pa.....	5/25/01			7/1		
12.	Cleveland, Ohio.....	5/25/01	6/15	21 da.	6/19	4	
13.	Cleveland, Ohio.....	6/5/01	6/25	20 da.	7/4	19	
14.	Philadelphia, Pa.....	6/7/01	6/27	20 da.	6/29	1½	Wilson
15.	Cleveland, Ohio.....	6/16/01	7/15	29 da.	7/20	5	
16.	Philadelphia, Pa.....	9/5/01	9/19	14 da.	9/24	5	
17.	Massachusetts.....	9/10/01	9/10	21 da.	Recov.		Wilson
18.	Spring City, Pa.....				9/19		
19.	Camden, N. J.....	10/1/01		7 wk.		7	Wilson
20.	Burlington, Vt.....	10/2/01	10/21	19 da.	Death	2	Wilson
21.	Camden, N. J.....	Fall '01					Wilson
22.	Burlington, Vt.....	Oct. '01		21 da.	Death		Wilson
23.	Philadelphia, Pa.....	10/9/01	11/6	28 da.	11/8	1½	Wilson
24.	Camden, N. J.....	10/14/01	11/8	25 da.	11/15	7	Wilson
25.	Philadelphia, Pa.....	10/14/01	11/4	21 da.	11/8	4	
26.	Camden, N. J.....	10/15/01?	11/1	17 da.			
27.	Atlantic City, N. J.....	10/18/01	11/8	21 da.			
28.	Camden, N. J.....	10/19/01	11/4	16 da.			
29.	Camden, N. J.....	10/19/01	11/6	18 da.	11/8	2	
30.	Philadelphia, Pa.....	10/19/01	11/10	22 da.	12/1	9	Wilson
31.	Bristol, Pa.....	10/19/01	11/10	22 da.	11/12	2	Wilson
32.	Camden, N. J.....	10/21/01	11/11	21 da.	11/14	3	
33.	Camden, N. J.....	10/23/01?	11/11	19 da.	11/13	2	
34.	Atlantic City, N. J.....	10/23/01	11/13	21 da.			
35.	Camden, N. J.....	10/25/01?	11/15	21 da.			
36.	Nova Scotia, Can.....	10/25/01	11/18	24 da.			
37.	Philadelphia, Pa.....	10/26/01	11/13	18 da.	Recov.		Wilson
38.	Camden, N. J.....	10/26/01					
39.	Philadelphia, Pa.....	10/27/01	11/11	16 da.	11/21	5	Wilson
40.	Atlantic City, N. J.....	10/28/01	11/16	19 da.			

CHART 1.—Continued.

	Location.	Date of vaccination.	Date of onset of tetanus.	No. of days between vaccination and onset.	Date of death.	No. of days between onset and death.	Reported by.
41.	Merchantville, N. J.	10/28/01					
42.	Bristol, Pa.	11/3/01	11/24	21 da.	11/27	3	Wilson
43.	Brighton, N. J.	11/4/01	11/27	23 da.	12/2	5	Wilson
44.	Camden, N. J.	11/4/01	11/25	21 da.	11/27	1½	Wilson
45.	Philadelphia, Pa.	11/8/01	11/27	19 da.	11/30	3	Hughes
46.	Philadelphia, Pa.	11/11/01	12/1	23 da.	12/16	15	Hughes
47.	Philadelphia, Pa.	11/18/01	11/25	7 da.	11/26	1	Hughes
48.	Philadelphia, Pa.	12/9/01	12/30	21 da.	Recov.		
49.	Montreal, Canada.	9/7/06	9/26	19 da.			
50.	San Francisco, Cal.	7/27/09	8/21	25 da.	8/26	1	
51.	Philadelphia, Pa.	8/20/09	9/17	28 da.	9/19	2	Wadsworth
52.	Bethlehem, Pa.	8/28/09?	10/14	47 da.	10/15	1	
53.	Hartford, Conn.	10/2/09	10/23	21 da.	10/25	2	
54.	W. Summerville, Mass.	1909			10/22		
55.	Philadelphia, Pa.	9/6/09	9/13	37 da.	9/15	2	Wadsworth
56.	Philadelphia, Pa.	9/9/09	10/5	26 da.	10/10	5	Wadsworth
57.	Lancaster, Pa.	8/7/10			10/23		Dr. Miller
58.	Lancaster, Pa.	8/24/10			10/9		Dr. Miller
59.	Philadelphia, Pa.	9/7/10	10/1	24 da.	10/5	4	Wadsworth
60.	St. Louis, Mo.	9/20/10			10/9		
61.	Philadelphia, Pa.	9/21/10	10/13	22 da.	10/16	3	Wadsworth
62.	St. Louis, Mo.	10/4/10	10/16	12 da.	10/17	1	
63.	St. Louis, Mo.	10/25/10					
64.	Philadelphia, Pa.	10/26/10	11/13	18 da.	11/15	2	Wadsworth
65.	Philadelphia, Pa.	11/5/10			12/4		Wadsworth
66.	Worrell Park, Md.	1911					
67.	Mt. Winans, Md.	1911					
68.	York Road, Md.	1911					
69.	Annapolis, Md.	1911			10/2		
70.	Philadelphia, Pa.	8/5/11	10/12	58 da.	10/14	2	Wadsworth
71.	Philadelphia, Pa.	8/28/11	9/22	25 da.	9/24	2	Wadsworth
72.	Philadelphia, Pa.	9/5/11			9/20		Wadsworth
73.	Philadelphia, Pa.	7/17/12	8/17	31 da.	8/21	4	Wadsworth
74.	Philadelphia, Pa.	9/1/12	9/16	15 da.	9/17	1	Wadsworth
75.	Philadelphia, Pa.	5/2/13	5/21	19 da.	5/22	1	Wadsworth
76.	Philadelphia, Pa.	5/14/13	6/2	19 da.	6/3	1	Wadsworth
77.	Philadelphia, Pa.	9/2/13	9/22	20 da.	9/24	2	Wadsworth
78.	Philadelphia, Pa.	9/8/13	9/22	14 da.	9/24	2	Wadsworth
79.	Gloversville, N. Y.	7/26/14	8/24	29 da.	8/27	3	
80.	Gloversville, N. Y.	8/11/14	9/6	26 da.	9/11	5	
81.	White Plain, N. J.	8/17/14	9/14	28 da.	Recov.		
82.	Aurora, N. Y.	8/24/14	9/9	16 da.	9/18	9	

This again would indicate infection subsequent to the vaccination.

The geographical distribution of these cases seems worthy of consideration. Of the 82 cases, 34 are from Pennsylvania, 16 from New Jersey, 6 from Ohio and 5 from New York; with the balance, 21, scattered over 9 other states and Canada. This in a measure conforms to a list of Andrews and Morgan of 1,099 cases of tetanus from all causes; Pennsylvania leading with 224 cases and the four states of Pennsylvania, Ohio, New York and New Jersey contributing 515 or nearly one half the total number reported by them in the United States.

Some critics have suggested that if tetanus following vaccination is simply a secondary infection, why are not more cases of tetanus reported as resulting from open wounds present at all times in large cities. In order to show that this disease does occur in localities where tetanus is known to be present in the dust, I will call attention to Doctor Wadsworth's article taken from the coroner's records of Philadelphia as follows:

1909

Total virus distributed by health department.....	40,400
Total cases of tetanus investigated.....	25
Post vaccinal.....	3

1910

Total virus distributed by health department.....	65,000
Total cases of tetanus reported.....	30
Post vaccinal.....	4

1911

Total virus distributed by health department.....	44,183
Total cases of tetanus reported.....	24
Post vaccinal.....	3

1912

Total virus distributed by health department.....	40,844
Total cases of tetanus reported.....	4
Post vaccinal.....	2

In these four years, the city distributed 190,427 vaccinations and Doctor Wadsworth estimates that private firms sold at least double this amount, making more than half a million vaccinations in the city. Tetanus followed in only 13 cases, while there were 90 cases of tetanus having no relation to vaccination. This would seem to demonstrate that certainly most cases of post-vaccinal tetanus can be accounted for in the same way as tetanus following any other wound presenting proper conditions for the reception and development of tetanus spores.

I will not take up your time in discussing the possibility of contaminated virus but will only call attention to the fact that if it were possible for tetanus bacilli to escape detection when the virus is subjected to the rigid tests required by the Public Health Service, Doctor Francis* recent work seems to indicate that the number of organisms must be so few that no harm could result. This is in line with Carini's† earlier work in which he detected tetanus bacilli five times in 400 experiments with 50 difficult lymphs. He states: "The presence of so small a number of tetanus germs forms no particular danger." and that "Some of the particular specimens of lymph in which we could demonstrate tetanus spores were used in thousands of inoculations without any complications resulting therefrom, whatever. Experi-

* Hygienic Laboratory.—Bulletin No. 95.

† Doctor Carini, formerly Chief of Vaccine Department, Bourne, Switzerland.

mentally it is only when the organism is present in numbers so large that they could be easily detected by cultural tests that we may fear the development of tetanus through contaminated virus."

I have thus far attempted to show that in the majority of post-vaccinal tetanus cases the evidence indicated late infection. But admitting this to be true, what can be done in the way of prevention? I would give the following suggestions:

1. The proper cleansing of the operation field.

2. Make proper scarification; short linear incision according to Dock's* method.

3. Instruction of the patient or parents as to after treatment. The vaccinator to examine the vaccination weekly or more often if necessary until the patient is well.

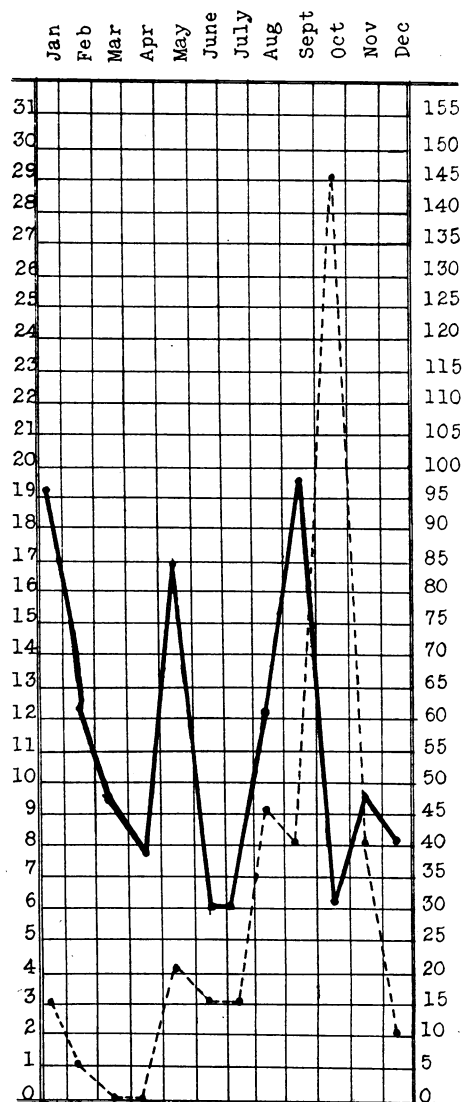
4. On evidence of infection of any kind, treat *promptly* as an infected wound, without regard to the vaccination.

5. Instruct the patient or parents as to the possibilities from neglect of these instructions.

6. Vaccinate only in the winter or spring unless smallpox is present. The emphasizing of this point is the most important object of this study.

In 77 of these cases, the date of onset of tetanus is known, and of this number, 29 developed tetanus in October, 15 in September, 10 in August and 8 in November. The other eight months of the year show a total of 15 cases with none in March or April.

CASES FROM POST-VACCINAL TETANUS COMPARED WITH OUTPUT OF VIRUS.



The black line indicates the average output of virus for the month.

The dotted line indicates the number of cases of post-vaccinal tetanus cases during the same month.

* George Dock A. M., M. D. Simple incisions for vaccinations. Therapeutic notes, Nov., 1906.

To summarize, therefore, over 80 per cent. of these cases occurred in the late summer and the three fall months. You would probably expect this because of the school vaccinations at this time, considering that this would represent the largest total vaccinations in the year. This, however is not the case. For 15 years, I have noted the output of vaccine from the laboratory is usually heavier in the winter than during the school vaccination period. A reference to the chart shows the number of cases of tetanus by months as compared with the output of virus for corresponding periods during the years 1912-1913-1914.

It will be noted that a large proportion of the tetanus cases develop in the fall, while large amounts of virus are also distributed in winter and spring with small case rate for tetanus. Does this not indicate that weather condi-

tions, dust, etc., are more favorable to the development of tetanus in the summer and fall than at other times?

In conclusion, therefore, I cannot too strongly recommend that health officers and others responsible for the vaccination of school children insist on a change in the time when this operation is performed, as a routine procedure.

RÉSUMÉ.

1. Tetanus is the most important complication of vaccination.

2. The evidence indicates that infection is, in a large majority of cases, post vaccinal.

3. It may be largely prevented (1) by proper care of the vaccinal wound and (2) by requiring certificates of vaccination from school children in the winter or spring rather than in the fall as at present.